Article

The Effects of Climate Change (normal values/anomalous values) on Fire Duration, in Greece Areas.

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Abstract

In this study, we examine and highlight the effects of climate change on forest fires duration that have afflicted Greece over the past 25 years (2000- 2025). Thus, we selected three distinct regions,of Greece, spanning a distance of approximately 1.000 kilometers, in order to ensure impartial findings that are not confined to a single area and, consequently to avoid creating a misleading impression. The first area of interest is Attica, with 2.100 rows of climatological data and fire duration records. The second area lies further south and is Crete, with 2.893 rows of climatological data and fire duration records. The third area of interest is located further north and is Eastern Macedonia & Thrace, with 2.919 rows of climatological data and fire duration records. The baseline for our analysis was established using climatological data from 1980–2010 for the regions of Attica and Crete, and from 1992–2022 for Eastern Macedonia and Thrace, reflecting the earliest available records for that area. These datasets were processed to derive monthly mean values representing the climatological normal's. We then compared these normal's with the meteorological conditions present at the onset of wildfire events during the period 2000–2025. Our findings report the climatological anomalous (climate change) influence of: temperature, wind speed, and humidity on wildfire duration.

Keywords: Climate change; machine learning; neural networks

1. Introduction

2. Materials and Methods

3. Results

To evaluate the predictive performance of the proposed feature construction method against widely used machine learning approaches, five models were trained and tested on the climatological and wildfire duration records of the three study regions: a genetic-neuroevolution network (NEAT), a radial basis function network (RBF), a feedforward neural network trained with the Adam optimizer, a feedforward neural network trained with the L-BFGS algorithm, and the proposed feature construction method based on grammatical evolution (FC2). Table 1 reports the mean relative error obtained by each method for Attica, Crete, and Eastern Macedonia & Thrace, together with the average performance across all three regions.

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Table 1. Experimental results on the used datasets using a series of machine learning methods.

DATASET	NEAT	RBF	ADAM	LBFGS	FC2
ATTICA	15.35%	4.81%	3.83%	2.98%	1.90%
CRETE	16.36%	1.79%	2.25%	3.70%	2.18%
MACEDONIA	14.24%	5.94%	7.56%	0.51%	0.46%
AVERAGE	15.32%	4.18%	4.55%	2.40%	1.51%

The differences among the five methods, summarized numerically in Table 1, are visualized in Figure 1 to facilitate a direct region-by-region comparison. The chart highlights that FC2 consistently ranks among the best-performing methods in Attica and Eastern Macedonia & Thrace, whereas in Crete the RBF network marginally outperforms it. NEAT, in contrast, exhibits substantially higher error across all three regions, underlining its limited suitability for this prediction task.

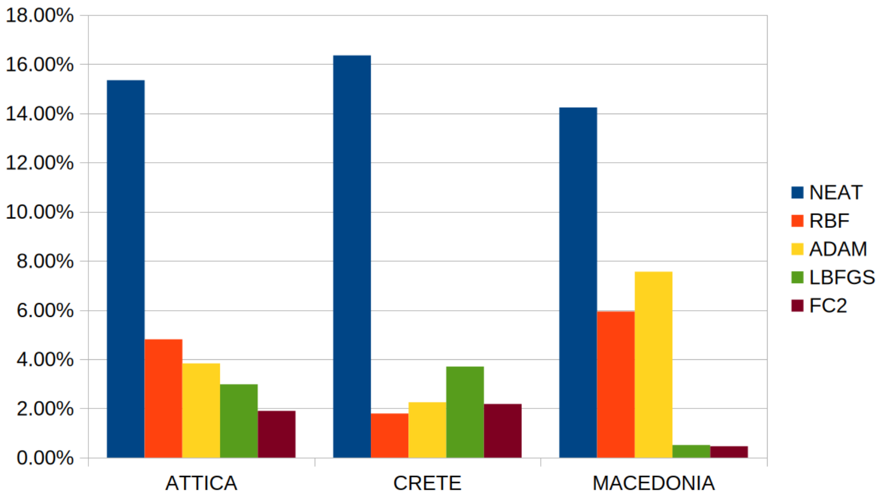

Figure 1. Graphical representation of the experimental results obtained by the application of various machine learning methods.

Beyond the comparison against alternative methods, we further examined how the number of features constructed by the grammatical evolution process affects model accuracy. Figure 2 illustrates the average error obtained when one, two, or three constructed features ($F = 1, F = 2, F = 3$) are used as inputs. Across all three regions, the error decreases as additional constructed features are introduced, with the most pronounced improvement observed in Eastern Macedonia & Thrace, indicating that extracting more than a single derived feature contributes measurably to the model's predictive capacity.

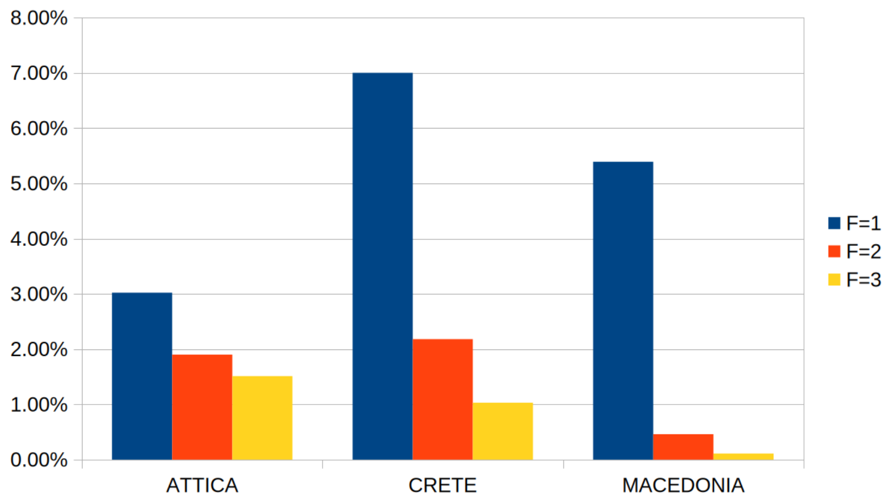

Figure 2. Graphical representation of the average classification obtained by the feature construction method using a variety of values for the number of constructed features $(F = 1, F = 2, F = 3)$.

The results of the statistical analysis are presented graphically in Figure 3, where the mean error and standard deviation of each method are shown alongside the corresponding significance level relative to FC2. Although the Friedman test did not reach the conventional 0.05 significance threshold (chi-squared = 8.533, p = 0.074), the Nemenyi critical difference of 3.522 confirms that the average rank of FC2 differs significantly from that of NEAT, consistent with the paired t-test result (p = 0.0002). The remaining comparisons, against LBFGS, RBF, and ADAM, do not reach statistical significance, which is likely attributable to the small number of regions (n = 3) available for the paired tests.

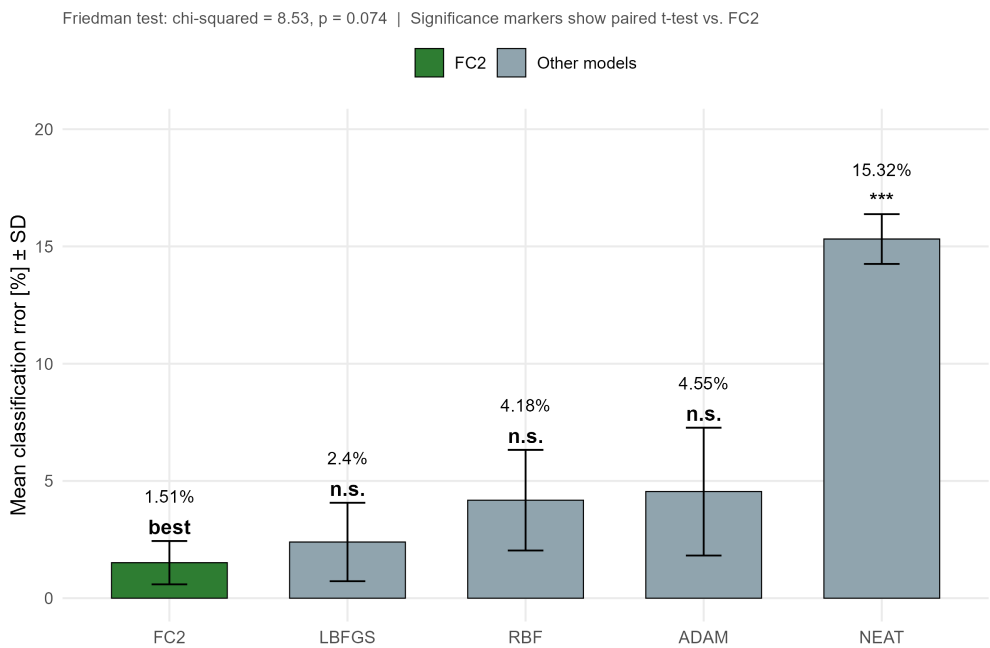

Figure 3. Mean prediction error per method with standard deviation and statistical significance relative to FC2.

To determine whether the observed differences in predictive accuracy are statistically meaningful rather than attributable to chance, a non-parametric Friedman test was applied across the three regions, followed by pairwise comparisons of FC2 against each of the remaining four methods using the paired Wilcoxon signed-rank test and the paired t-test.

Model	Mean Error	SD Error	Min Error	Max Error	Avg Rank	Mean Diff vs FC2	Wilcoxon p	Paired t_p	Significance
FC2	1.51	0.92	0.46	2.18	1.33	0			Reference
LBFGS	2.4	1.67	0.51	3.7	2.67	-0.88	0.1814	0.1798	n.s.
RBF	4.18	2.15	1.79	5.94	2.67	-2.67	0.4227	0.2571	n.s.
ADAM	4.55	2.73	2.25	7.56	3.33	-3.03	0.1814	0.286	n.s.
NEAT	15.32	1.06	14.24	16.36	5	-13.8	0.1814	0.0002	***

Table 2. Statistical comparison of the proposed FC2 method against NEAT, RBF, ADAM, and LBFGS.

Table 2 summarizes the descriptive statistics, average ranks, and significance outcomes of these comparisons, using FC2 as the reference method.

Friedman test: chi-squared = 8.533, df = 4, p-value = 0.0739

Nemenyi critical difference (alpha = 0.05): 3.522

*Significance vs. FC2 (paired t-test): *** p<0.001, ** p<0.01, * p<0.05, n.s. = not significant*

4. Conclusions

Author Contributions: For research articles with several authors, a short paragraph specifying their individual contributions must be provided. The following statements should be used "Conceptualization, X.X. and Y.Y.; methodology, X.X.; software, X.X.; validation, X.X., Y.Y. and Z.Z.; formal analysis, X.X.; investigation, X.X.; resources, X.X.; data curation, X.X.; writing—original draft preparation, X.X.; writing—review and editing, X.X.; visualization, X.X.; supervision, X.X.; project administration, X.X.; funding acquisition, Y.Y. All authors have read and agreed to the published version of the manuscript." Please turn to the CRediT taxonomy for the term explanation. Authorship must be limited to those who have contributed substantially to the work reported.

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